01 · SITUATION

A marketplace where most orders are fine and the failures are concentrated.

98,962
orders analysed, 86 weeks

90.5%
arrived on time

6.8%
late, of delivered orders

62.4%
rated 1–2★ when late

A late order is rare. When it happens the customer reaction is severe: 62.4% leave 1–2★, against 9.2% on an on-time order.

Eight weeks out of 86 were genuinely unusual, not just noisy.

8
weeks flagged SEV1 by the control chart

41.5%
of all late orders fall in those 8 weeks

2
separate episodes: Black Friday 2017, Feb–Mar 2018

A control chart on the weekly late rate, baselined on the median of the prior 26 weeks, so an alert means statistically unusual rather than simply higher than last week.

Same symptom, two different causes. Sellers were not the main one.

72.3%
of late orders had already left the seller on time

12.4%
would have been saved by an on-time handoff alone

42.2%
late rate in Rio de Janeiro, Feb–Mar 2018

Black Friday stretched both legs (1 days on the seller leg, 2.8 days on the carrier leg). February–March barely touched the seller leg (0.3 days) and added 4.1 days to the carrier leg. Chasing slow sellers would have missed the worse episode entirely.

04 · WHAT TO DO

Three moves, ranked by what the data actually supports.

1 Wire the monitor to a weekly review with a defined SEV1 trigger. 0–30 days

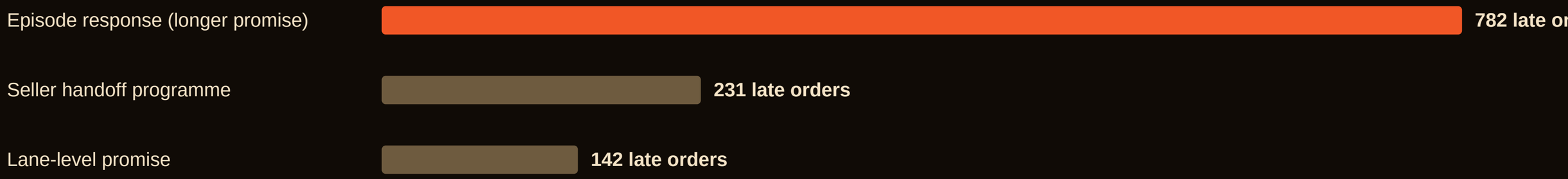
2 Run the episode-response playbook on the next flagged week. 30–60 days

3 Put the 89 flagged sellers on a handoff improvement track. 60–90 days

Owners shown on the website are illustrative functional roles. No Instacart team, tool or internal metric is named or estimated anywhere in this project.

05 · WHAT IT IS WORTH

Counting only what could actually have been acted on at the time.



Black Friday is on the calendar, so it is counted in full. Feb–Mar 2018 counts only orders placed after the monitor could have confirmed the episode, using the dataset's own median promise length as the detection delay. The cost in lost conversions from a longer promise cannot be measured here.

06 · LIMITS

What this evidence does not support.

— Multi-day parcel shipping in Brazil, 2017–2018. The methods transfer; the timescales and the specific numbers do not.

— No reschedule events and no shopper-supply data, so two defect types a real fulfilment team tracks cannot be shown.

— Fault attribution uses timestamps, not pickup scans: a seller delayed by a missed carrier pickup is blamed unfairly.

— The repeat-purchase result rests on 35 and 41 repeat customers. A consistent signal, not a measured effect size.

— The early-warning signal was tested and dropped: it gave no usable lead time before either episode.